

Object-Oriented Programming - Midterm (sample)

Typed exam script for RExA. Each question is followed by the student answer.

Question:

Explain the concept of encapsulation in object-oriented programming and describe how access modifiers support it.

Answer:

The main idea is that encapsulation bundles an object's data and behavior together while restricting direct access to its internal state. For example, access modifiers like private, protected, and public control which fields and methods can be seen from outside the class. This is because hiding the internal fields prevents other parts of the program from making the object enter an invalid state. For instance, a class can expose getter and setter methods instead of allowing direct access to a private field. This matters because the setter can validate incoming values before the internal state is ever changed. In conclusion, encapsulation and access modifiers work together to achieve data hiding and reduce coupling between classes.

Question:

Compare inheritance and polymorphism in object-oriented programming, with examples.

Answer:

The main idea is that inheritance is like sharing a family pizza recipe so a child class reuses the parent class. For example, a Dog class can inherit from Animal. Polymorphism is when different classes respond to the same method in their own way, therefore speak() can bark or meow. In conclusion, inheritance reuses code and polymorphism lets one interface have many behaviors.